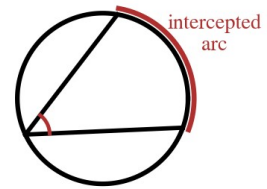


Speak Like A Mathematician:

inscribed angle:

intercepted arc:

inscribed polygon:



Inscribed Angle Theorems

Inscribed Angle Theorem: an inscribed angle measures _____ of its intercepted arc.

Two inscribed angles that intercept the SAME arc are _____ .

An angle inscribed in a semicircle is a _____ angle.

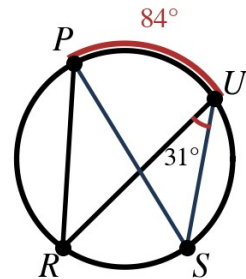
A quadrilateral can be inscribed in a circle if and only if its opposite angles are _____ .

Example 1 - Inscribed Angles in Circle

In the figure at the right, $m\widehat{PU} = 84^\circ$ and $m\angle RUS = 31^\circ$.

a) Find $m\angle PRU$.

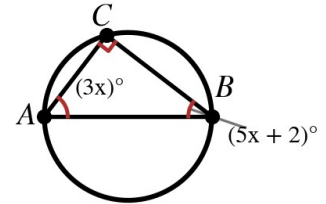
b) Find $m\widehat{RS}$.



Example 2 - Right Triangle in a Circle

AB is a diameter, so $\triangle ABC$ is inscribed in a semicircle. $m\angle A = (3x)^\circ$ and $m\angle B = (5x + 2)^\circ$.

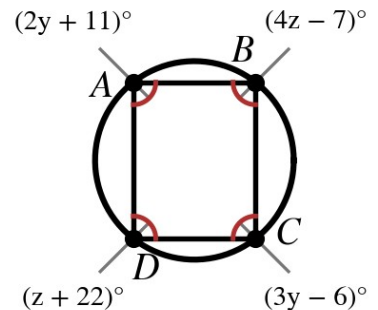
- Why is $\angle C$ a right angle?
- Solve for x , then find both acute angles.



Example 3 - Inscribed Quadrilateral

$ABCD$ is inscribed in the circle: $\angle A = (2y + 11)^\circ$, $\angle C = (3y - 6)^\circ$, $\angle B = (4z - 7)^\circ$, $\angle D = (z + 22)^\circ$.

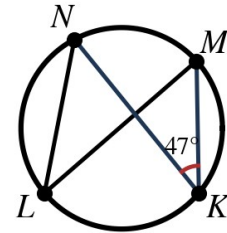
- Find y , then $\angle A$ and $\angle C$.
- Find z , then $\angle B$ and $\angle D$.



Example 4 - Same Arc, Same Measure

In the top figure, $\angle NKM = 47^\circ$.

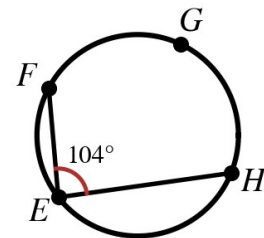
- a) Find $m\angle NLM$ and explain.



Example 5 - A Big Intercepted Arc

In the middle figure, inscribed $\angle FEH = 104^\circ$.

- a) Find $m\widehat{FGH}$.
- b) Why can an intercepted arc be bigger than 180° ?



Example 6 - Central vs. Inscribed

In the bottom figure, central $\angle LKM = 98^\circ$.

- a) Find $m\widehat{LM}$.
- b) Find inscribed $m\angle LJM$.

